

Atividade

Passo 4.6

print demonstra a realização de testes de autenticação utilizando a ferramenta **radtest** no Kali Linux para validar a comunicação com um servidor **RADIUS** (Remote Authentication Dial-In User Service).

```
}
Listening on auth address * port 1812 bound to server default
Listening on acct address * port 1813 bound to server default
Listening on auth address :: port 1812 bound to server default
Listening on acct address :: port 1813 bound to server default
Listening on auth address 127.0.0.1 port 18120 bound to server inner-tunnel
Listening on proxy address * port 49909
Listening on proxy address :: port 45259
radtest: ^C
cleydson@Cleydson: ~
Sessão  Ações  Editar  Exibir  Ajuda
(cleydson@Cleydson)-[~]
$ radtest usuariol rnpesr 127.0.0.1 0 testing123
sent Access-Request Id 185 from 0.0.0.0:42277 to 127.0.0.1:1812 length 78
  User-Name = "usuariol"
  User-Password = "rnpesr"
  NAS-IP-Address = 127.0.1.1
  NAS-Port = 0
  Message-Authenticator = 0x00
  Cleartext-Password = "rnpesr"
received Access-Accept Id 185 from 127.0.0.1:1812 to 127.0.0.1:42277 length 38
  Message-Authenticator = 0x4a8ae8fe05f9bf769de6fdf8cfd6f185
(cleydson@Cleydson)-[~]
$
```

```
(0) [preprocess] = ok
(0) [chap] = noop
(0) [mschap] = noop
(0) [digest] = noop
(0) suffix: Checking for suffix after "@"
(0) suffix: No '@' in User-Name = "usuariol", looking up realm NULL
(0) suffix: No such realm "NULL"
(0) [suffix] = noop
(0) eap: No EAP-Message, not doing EAP
(0) [eap] = noop
(0) File /etc/raddb/clients.conf: entry removed at line 207
```

Passo 4.7

print demonstra a implementação de **Autenticação de Dois Fatores (2FA)** no Linux utilizando o **Google Authenticator**, além da configuração dos módulos de autenticação plugáveis (**PAM**) e do

serviço **SSH** para reforçar a segurança de acesso remoto.

```
Your new secret key is: 4SLT3NA6FH63VA5PR63MIEZSPJQRGNXP
Enter code from app (-1 to skip): y
Code incorrect (correct code 530951). Try again.
Enter code from app (-1 to skip): 893302
Code confirmed
Your emergency scratch codes are:
  73897034
  76289253
  18333520
  74519796
  32588739

Do you want me to update your "/home/cleydson/.google_authenticator" file? (y/n) y

Do you want to disallow multiple uses of the same authentication
token? This restricts you to one login about every 30s, but it increases
your chances to notice or even prevent man-in-the-middle attacks (y/n) y

By default, a new token is generated every 30 seconds by the mobile app.
In order to compensate for possible time-skew between the client and the server,
we allow an extra token before and after the current time. This allows for a
time skew of up to 30 seconds between authentication server and client. If you
experience problems with poor time synchronization, you can increase the window
from its default size of 3 permitted codes (one previous code, the current
code, the next code) to 17 permitted codes (the 8 previous codes, the current
code, and the 8 next codes). This will permit for a time skew of up to 4 minutes
between client and server.
Do you want to do so? (y/n) y

If the computer that you are logging into isn't hardened against brute-force
login attempts, you can enable rate-limiting for the authentication module.
By default, this limits attackers to no more than 3 login attempts every 30s.
Do you want to enable rate-limiting? (y/n) y

(cleydson@cleydson)-[~]
$ sudo -i
(root@cleydson)-[~]
# sudo nano /etc/pam.d/common-auth

(root@cleydson)-[~]
# sudo nano /etc/pam.d/sshd

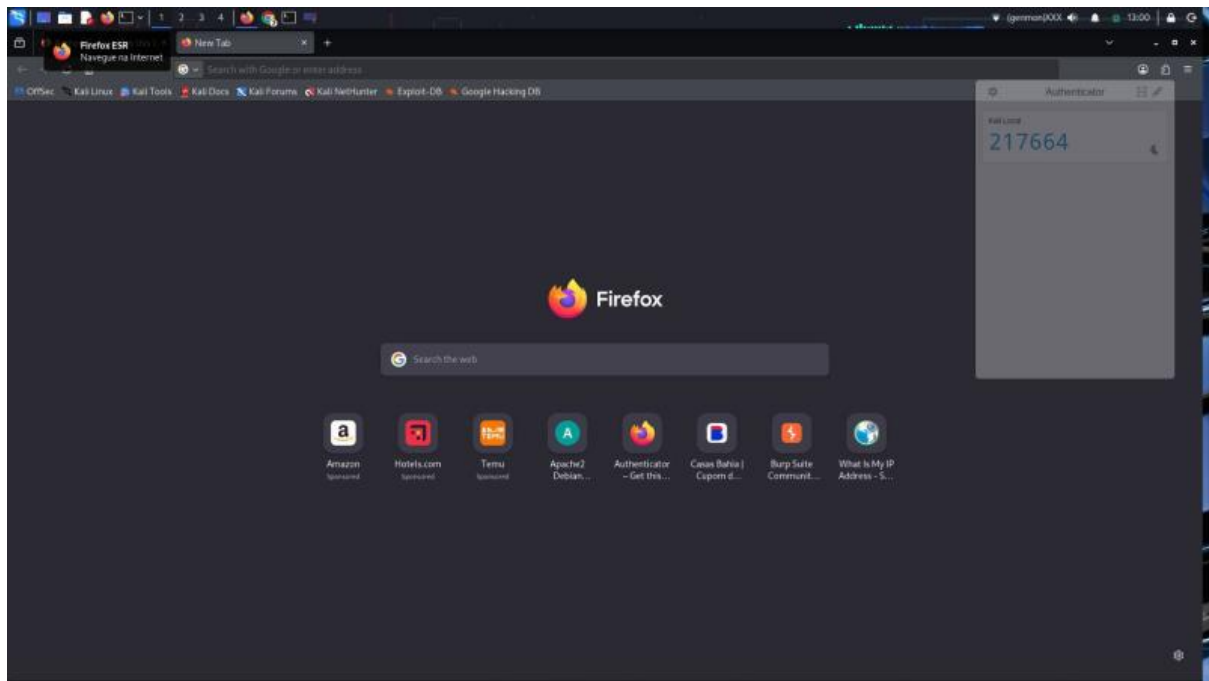
(root@cleydson)-[~]
# sudo nano /etc/ssh/sshd_config

(root@cleydson)-[~]
# sudo systemctl restart ssh

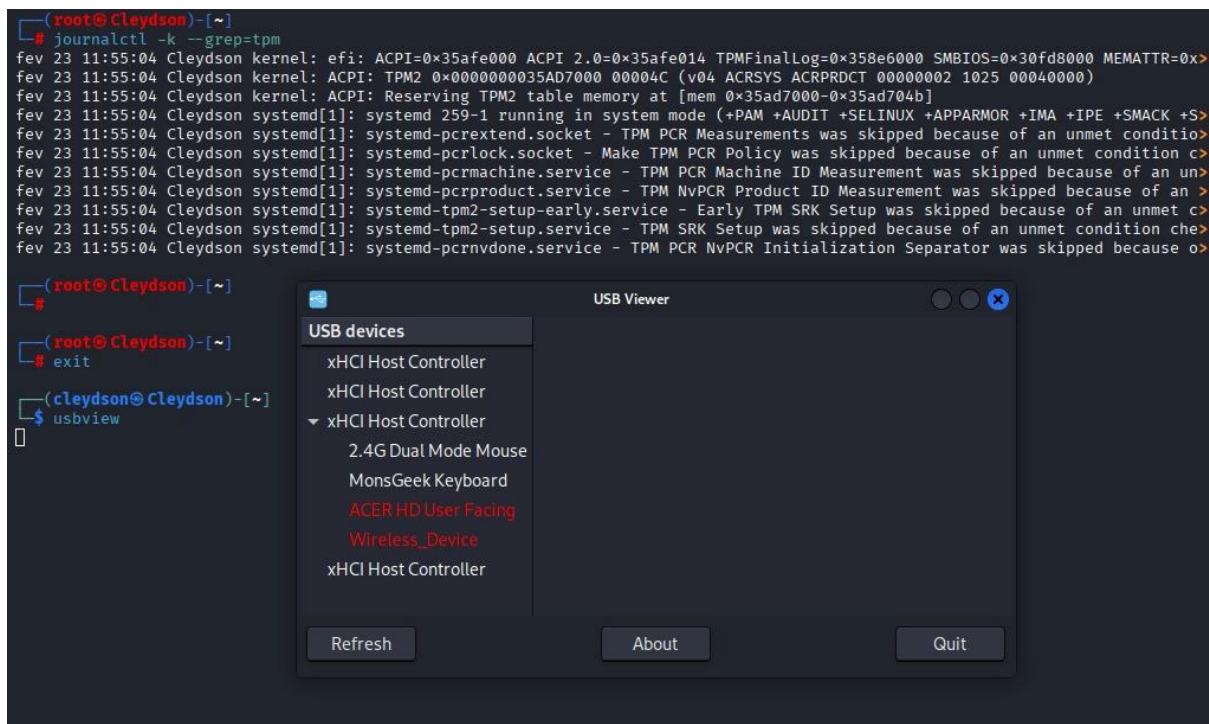
(root@cleydson)-[~]
#
```

Passo 4.8

prints demonstram a fase final de validação da **Autenticação de Dois Fatores (2FA)**. Neles, é possível ver o uso de uma aplicação autenticadora integrada ao navegador Firefox no Kali Linux para gerar os códigos temporários (TOTP) necessários para o acesso seguro.



Passo 4.9



prints demonstram a realização de auditorias de hardware no sistema Linux, focando na verificação de módulos de segurança física e periféricos conectados.

Passo 4.10

print demonstra a validação final do sistema de **Autenticação de Dois Fatores (2FA)** utilizando o **OTPClient**. A imagem destaca a geração de um código temporário (TOTP) especificamente configurado para o programa **Hackers do Bem**

